Calculus 2 Workshop

Little Hall Tutorial Program

Semester 1, 2026

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How to prepare

1. Make a4 notes sheet draft.
   1.1 Do a run through your notes.
   1.2 Calculus 2 is all patterns.
2. Look back at difficult tutorial questions.
3. Try past exam questions.
   3.1 Alba taught Sem. 1 2025, so look at that exam.

Be careful:

1. Don't waste your time
2. Timed practice exams can burn you out

What goes in your a4 notes? — Examples

1. Standard limits
2. 'If ... then ...' (calculus 2 is pattern recognition).
   - 2.1 Standard forms for limits/series (next slide). (See Slide 6.)
   - 2.2 What techniques to use where.
3. Graph shapes (especially the 'arc' graphs).
4. Local max/min criteria.
5. Partial fraction forms.
6. DE techniques
   - 6.1 How to find general or particular solutions.
   - 6.2 When are equilibria stable?
   - 6.3 'Homogeneous type' $\rightsquigarrow$ $u = y/x$ reduces to separable ODE.

Advice for the a4 notes

1. If you didn't use it in tutorials, it's probably too specific for the exam.
2. Don't write down everything (this is why you make a draft).
3. Include 'safety facts'. I liked putting the integrating factor and directional derivative formula in, even though I know them by heart.

Standard forms in Calculus 2

You should go through and make your own table of these standard forms and include some on your a4 notes sheet.

1. For integrals

$$\int \sqrt{x^2 + a^2}\,dx \rightsquigarrow \text{let } x = a\sinh(t)$$

$$\int \sqrt{x^2 - a^2}\,dx \rightsquigarrow \text{let } x = a\cosh(t).$$

2. For limits
   - 2.1 With functions of x, try use L'Hopital.
   - 2.2 A sum with many terms, divide through by fastest growing

$$\lim_{n\to\infty} \frac{5^n + n!}{2^n + n!} = \lim_{n\to\infty} \frac{\frac{5^n}{n!} + 1}{\frac{2^n}{n!} + 1} = \frac{0+1}{0+1} = 1.$$

3 For series

3.1 Look at fastest growing terms in numerator and denominator to make an educated guess about divergence or convergence

$$\frac{3n^2+4n}{n^3+5n^2} \approx \frac{3n^2}{n^3} = \frac{3}{n}$$

so you might expect

$$\sum_{n=1}^{\infty} \frac{3n^2+4n}{n^3+5n^2}$$

to diverge.

3.2 Then try comparison test

$$\sum_{n=1}^{\infty} \frac{3n^2+4n}{n^3+5n^2} \geq \sum_{n=1}^{\infty} \frac{3n^2}{n^3+5n^2} \geq \sum_{n=1}^{\infty} \frac{3n^2}{n^3+5n^3} = \sum_{n=1}^{\infty} \frac{1}{2n} = \infty.$$

3.3 With lots of factorials and powers, use ratio test.

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}2^n n!}{(2n)!}.$$

Exam strategies

1. Look for easy marks at the start.
2. Skipping around is great.
   2.1 Use your formula sheet to mark skips.
3. Understand who is marking your exam.
4. Draw pictures where possible.
5. The last 5 minutes:
   5.1 Write down key words or theorems.
   'Divergence test', 'use integrating factor'.
   5.2 Guess the rapid-fire questions at the start of the exam. Some of them will be challenging.

Practice questions

1. Evaluate

$$\lim_{n\to\infty} \frac{(-1)^n n}{1+n^5}.$$

2. Do the following converge or diverge?

$$\sum_{n=1}^{\infty} \frac{3n^2+4n+n\log n}{n^3+5n+\cos^2(n)}, \quad \sum_{n=1}^{\infty} \frac{(n^2)!}{n!n!} \text{ (harder).}$$

3. Solve the DE

$$\frac{dy}{dx} + xy = e^{x^2}, \quad y(0) = 1.$$

4. Evaluate

$$\int \sqrt{4x^2+3}dx.$$

Where can I get these slides?



Available after tonight.